Lead Service Line Market Recommendation

Executive summary | Xylem / IBM SkillsBuild project

Xylem should use state inventory patterns to identify candidate markets for tailored identification and replacement-planning services, then assess individual utilities before selecting pilots. This approach links the type of reported infrastructure work to a proposed service emphasis.

Evidence supporting the recommendation

1. Identification is the larger reported inventory. The analytical sample contains 21.59 million Unknown lines and 3.12 million known replacement lines, a 6.9-to-1 ratio. Unknown inventory is larger in 49 of 51 jurisdictions. These counts describe lines, not relative costs or effort.

2. The inventory mix differs by state. Texas and Pennsylvania have the largest Unknown inventories (2.18 million and 2.06 million). Illinois leads known replacement inventory (672,210 lines). Ohio and New York have substantial inventories on both measures.

3. The measurement lens changes comparisons. Only 3 of the raw top 10 Unknown-inventory states and 6 of the known-replacement top 10 remain in their respective top 10 after population adjustment. Reporting also varies: New Hampshire contributes just 3 of 706 listed systems.

Proposed service emphasis and next action

Investigate identification support in Texas and Pennsylvania, replacement planning in Illinois, and parallel work in Ohio and New York. These are illustrative candidates. Each still has work on both dimensions, and other states may also warrant investigation.

Xylem's published materials describe detection, inventory development, funding assistance and replacement planning. Confirm current service scope and utility needs before a pilot. Review local material records, GIS, funding and delivery capacity, then agree on a baseline and measures such as classification progress, cost per verified line and progress toward actionable replacement plans.

Scope and decision limits

The analysis covers 43,993 complete-report Community Water Systems across 50 states and DC. Known Replacement Inventory means Lead plus Galvanized Requiring Replacement. Unknown does not mean lead. System reporting coverage is not service-line coverage; population-adjusted values are not affected residents. The data do not establish customer demand, project readiness, exposure or water quality.

Next question: Which utilities in the candidate markets have the inventory needs, data, funding, and capacity to support a pilot?

Sources EPA 2026 Q2 inventory | Census 2025 population | Xylem 2022 brochure pp 4–5 | Xylem asset management
Xylem sources reviewed October 11, 2026. Data, definitions and calculations accompany this summary in the notebook.